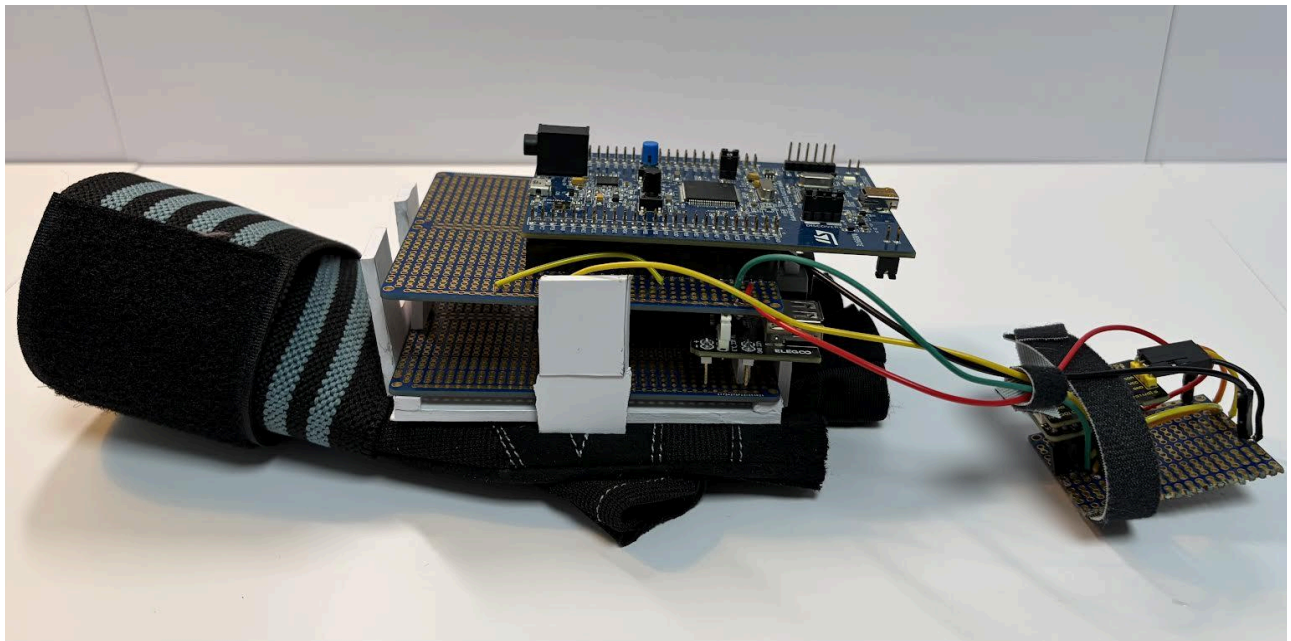


A Motion-Aware Embedded Heart-Rate Monitoring Framework Using Three-Mode NLMS Adaptive Filtering, Frequency-Domain Estimation, and Signal Quality Assessment



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Personal Engineering Project

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Abstract

Heart-rate monitoring using photoplethysmography (PPG) has become widely adopted in wearable devices due to its low cost, non-invasive nature, and ease of integration. However, PPG signals are highly susceptible to motion artifacts, which can significantly degrade heart-rate estimation accuracy during daily activities. This project presents a motion-aware heart-rate monitoring framework that combines photoplethysmography (PPG) and inertial measurement unit (IMU) sensing to improve robustness under motion conditions. The proposed system utilizes a MAX30102 PPG sensor, an LSM6DS3 IMU sensor, an STM32F407 microcontroller for real-time signal processing, and an ESP32 module for wireless data transmission and visualization.

The proposed framework incorporates multiple stages for motion-robust heart-rate estimation, including signal preprocessing, motion-aware adaptive noise cancellation, frequency-domain heart-rate estimation, signal quality assessment, and decision logic. A three-mode Normalized Least Mean Squares (NLMS) adaptive filtering strategy is introduced, where the filter dynamically operates in BYPASS, ADAPT, or FREEZE mode according to the detected motion level. This approach enables adaptive noise cancellation to be selectively applied while avoiding unnecessary signal distortion during low-motion conditions and unstable coefficient adaptation during severe motion.

Experimental evaluation was performed under no-motion, 1 Hz, 2 Hz, and 3 Hz hand-swinging conditions using an external pulse oximeter as a reference device. A correlation-based reference-signal selection study identified the X-axis acceleration signal as the most suitable adaptive filtering reference. Across all motion conditions, the proposed framework achieved mean absolute errors of 0.71 BPM, 1.08 BPM, 1.14 BPM, and 1.19 BPM, respectively. Under the most challenging 3 Hz motion condition, the complete processing framework reduced the mean absolute error from 5.48 BPM to 1.19 BPM and the root mean square error from 19.73 BPM to 1.41 BPM, corresponding to improvements of 78.3% and 92.9%, respectively. The results further indicate that while adaptive noise cancellation can provide benefits under severe motion conditions, the largest improvements in estimation accuracy and output stability were achieved through the combined effects of signal preprocessing, robust frequency-domain heart-rate estimation, signal quality assessment, and hold-mode decision logic. These results demonstrate the effectiveness of the proposed motion-aware processing framework for maintaining accurate and stable heart-rate estimation under varying motion intensities.

1.Introduction

1.1 Background

Heart rate is an important physiological parameter that provides insight into cardiovascular activity, physical exertion, and overall health status. The growing popularity of wearable health-monitoring devices has increased demand for compact, low-cost, and non-invasive methods of continuously measuring heart rate during daily activities. Photoplethysmography (PPG) is one of the most widely used techniques for wearable heart-rate monitoring. PPG uses an optical sensor to measure changes in blood volume within tissue, allowing heart rate to be estimated from the resulting waveform. Due to its low cost, low power consumption, and ease of integration, PPG has become the primary sensing technology in many smartwatches and fitness trackers. Despite its widespread use, PPG signals are highly susceptible to motion artifacts caused by body movement and variations in sensor–skin contact. These disturbances can distort the measured waveform and reduce the accuracy of heart-rate estimation, making motion robustness a key challenge in wearable physiological monitoring systems.

1.2 Problem Statement

While PPG is widely used for wearable heart-rate monitoring, obtaining accurate measurements during movement remains a significant challenge. In real-world use, people rarely remain stationary while wearing a device. Everyday activities such as walking, exercising, or simply moving the hand can introduce disturbances into the PPG signal that are unrelated to the actual heartbeat. A major cause of these disturbances is sensor–skin interaction. As the sensor shifts slightly on the skin or the contact pressure changes, the amount of light detected by the sensor can vary independently of blood flow. These unwanted variations can distort the PPG waveform and make it difficult to distinguish true cardiac activity from motion-induced noise. As a result, conventional heart-rate estimation methods may produce inaccurate or unstable measurements during motion. Improving the reliability of heart-rate monitoring under these conditions is therefore an important challenge for wearable devices. This project addresses this problem by combining PPG and IMU sensing with motion-aware signal processing techniques designed to reduce the impact of motion artifacts and improve heart-rate estimation.

1.3 Project Objectives

The primary objective of this project is to develop a wearable heart-rate monitoring system capable of operating under both stationary and motion conditions. To achieve this goal, the project aims to:

- Develop a real-time PPG-based heart-rate monitoring system.
- Improve heart-rate estimation robustness in the presence of motion artifacts.
- Utilize IMU measurements to characterize motion and support artifact reduction.
- Implement all signal processing algorithms on an STM32 microcontroller.
- Evaluate system performance under different motion conditions using a reference heart-rate device.

1.4 Contributions

The main contributions of this project are summarized as follows:

- Development of a motion-aware heart-rate monitoring system that integrates PPG and IMU sensing for robust heart-rate estimation under motion conditions.
- Design and implementation of a motion-aware adaptive noise cancellation framework using IMU-referenced NLMS filtering for motion artifact suppression.
- Introduction of a three-mode adaptive filtering strategy consisting of **BYPASS**, **ADAPT**, and **FREEZE** modes, enabling the adaptive filter to dynamically respond to varying motion intensities.
- Development and experimental validation of a correlation-based reference signal selection method, resulting in the selection of the X-axis acceleration signal as the optimal NLMS reference input.
- Implementation of an FFT-based heart-rate estimation algorithm incorporating spectral averaging, peak validation, continuity-based tracking, and confidence estimation.
- Development of a signal quality assessment and decision framework that evaluates estimate reliability and utilizes hold-mode operation to reject unreliable heart-rate estimates.
- Real-time implementation and validation of the complete processing pipeline on an STM32F407 microcontroller operating at a sampling frequency of 100 Hz.
- Experimental evaluation demonstrating accurate heart-rate estimation under no-motion, 1 Hz, 2 Hz, and 3 Hz motion conditions, with significant improvements in accuracy and output stability achieved through signal quality assessment and hold-mode processing.

2.System Overview

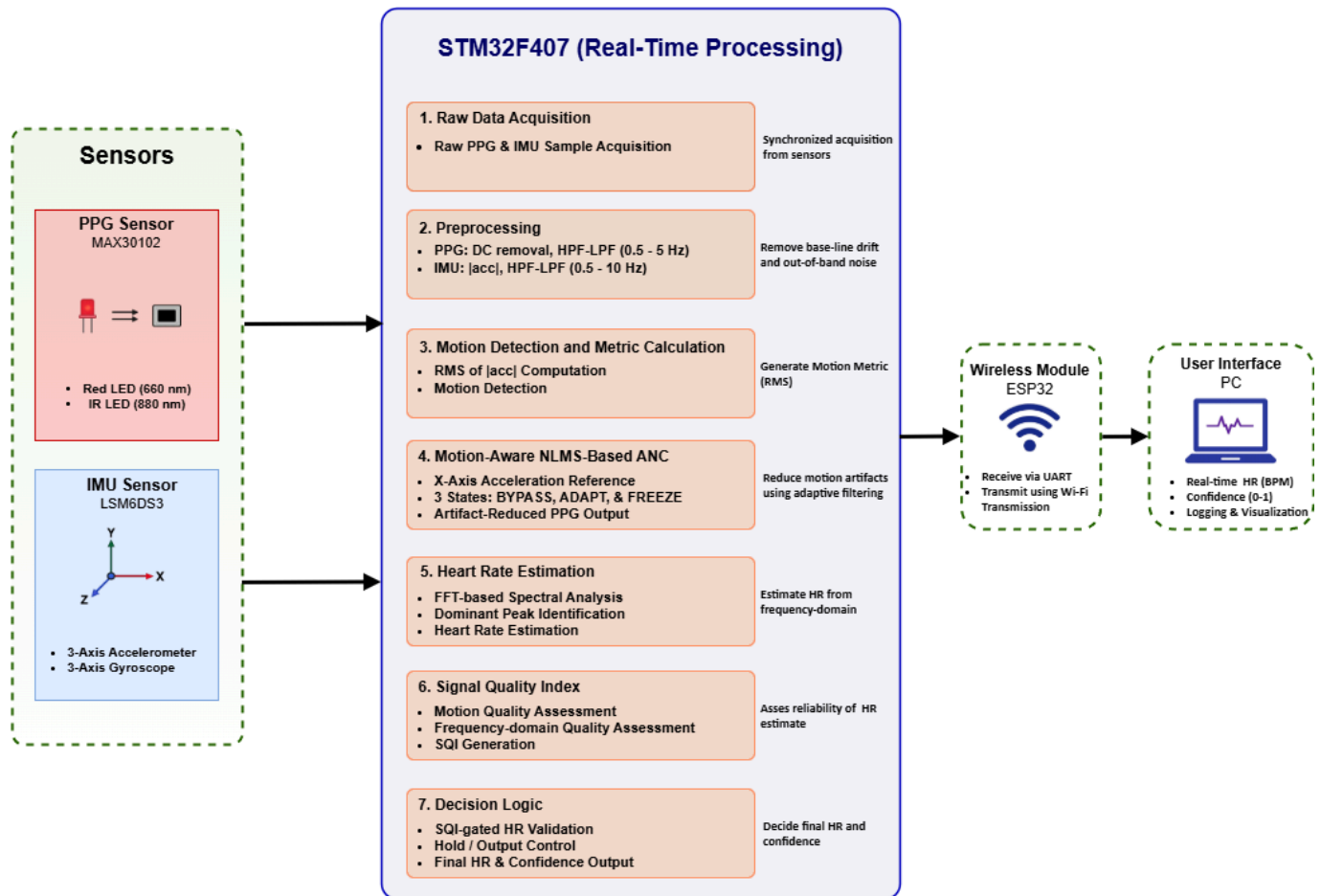


Figure 1: System Overview

Shown in *Figure 1* is a system overview, displaying the hardware architecture and the overall data flow. The project displays an architecture that incorporates photoplethysmography (PPG) and inertial measurement unit (IMU) sensing to allow an accurate heart rate estimation under motion conditions. The hardware consists of a MAX30102 PPG sensor, an LSM6DS3 IMU sensor, a STM32F407 microcontroller for real-time sampling and signal processing, along with an ESP32 Wi-Fi module for wireless data transmission to a PC for real-time data analysis and visualization.

The MAX30102 acquires optical PPG signals through integrated red and infrared LEDs and a photodetector, and the LSM6DS3 acquires user's motion data through its incorporated 3-axis

accelerometer and 3-axis gyroscope. The sensors communicate with the MCU through the I2C interface. The acquired sensor data is processed real-time in the MCU through a multi-stage signal processing pipeline, which consists of data acquisition, signal preprocessing, motion metric computation, NLMS-based adaptive noise cancellation, heart rate estimation, signal quality index(SQI) assessment, and decision logic.

The resulting heart rate estimate and confidence metrics(0-1) are transmitted from the MCU to the ESP32 Wi-Fi module through the UART interface. The ESP32 allows wireless communication with the PC for real-time visualization, logging, and monitoring on a host computer.

3. Signal Processing Pipeline

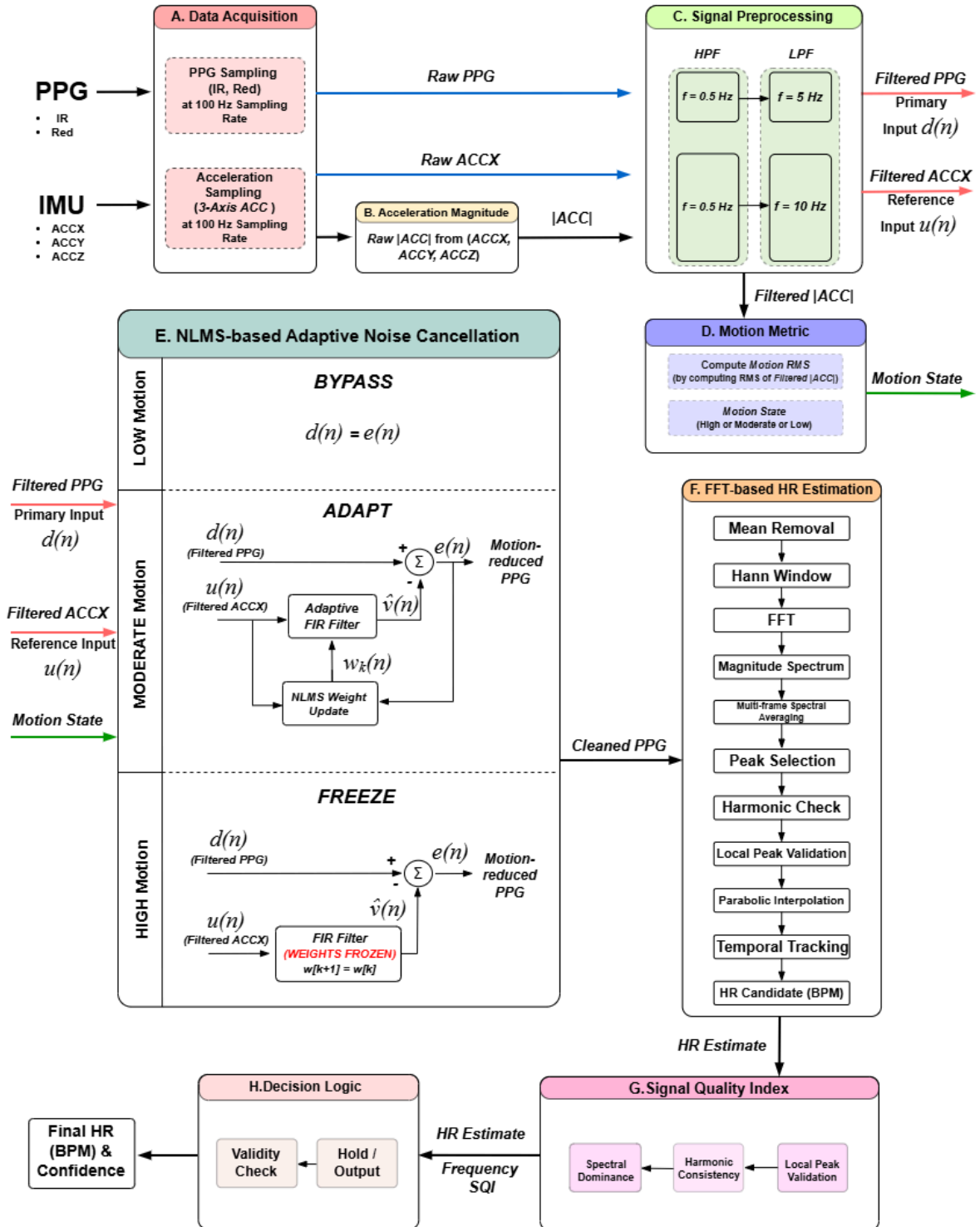


Figure 2: Signal Processing Pipeline

Figure 2 illustrates the signal processing flow of the proposed heart rate monitoring system, beginning with the acquisition of PPG and IMU data and ending with the generation of a heart rate estimate. The implemented signal processing pipeline consists of eight modular processing blocks, where each block performs a specific signal processing task and passes its output to the subsequent block. This modular architecture improves code organization, simplifies debugging, and allows individual processing stages to be developed and evaluated independently.

3.1 Block A: Sample Acquisition

In this block, PPG and IMU samples are acquired by the microcontroller at a sampling rate of 100 Hz. A periodic hardware timer interrupt triggers the acquisition process every 10 ms, ensuring a fixed sampling interval and consistent timing throughout the signal processing pipeline.

The PPG measurements are obtained from the MAX30102 optical sensor. The sensor consists of integrated red and infrared LEDs and a photodetector that measures variations in reflected light intensity caused by pulsatile blood volume changes within the tissue. For the MAX30102 sensor used in this project, the wavelengths of the red and infrared LEDs are 660 nm and 880 nm, respectively. The infrared channel is selected for subsequent heart rate estimation due to its stronger pulsatile component and improved penetration through biological tissue.

Motion information is acquired from the LSM6DS3 inertial measurement unit (IMU). The sensor provides three-axis acceleration measurements corresponding to motion along the X, Y, and Z directions. These measurements are used as a reference signal for motion artifact detection and suppression in later processing stages.

During each acquisition cycle, the MCU retrieves the latest sensor measurements through the I²C communication interface. The acquired samples are then stored in circular buffers together with sample validity information and sample indices. This buffering mechanism enables subsequent processing blocks to operate on fixed-length data windows while maintaining continuous real-time operation. Once both PPG and IMU samples have been successfully acquired and stored, the data are made available to the preprocessing stage for further signal conditioning.

3.2 Block B: Acceleration Magnitude Computation

The acquired acceleration samples are processed to compute an approximated acceleration magnitude from the three acceleration components. The acceleration magnitude, (a_{mag}), is calculated according to Equation x:

$$a_{mag} = |a_x| + |a_y| + |a_z|$$

where a_x , a_y , and a_z represent the acceleration measurements along the X, Y, and Z-axes, respectively.

Rather than relying on a single acceleration axis, combining all three axes provides a single measure of overall motion intensity and reduces the dependence on sensor orientation. Furthermore, the adopted formulation avoids the square-root operation required by the Euclidean magnitude calculation, resulting in a computationally efficient implementation suitable for real-time embedded processing on the STM32 microcontroller.

The computed acceleration magnitude is subsequently used for motion metric generation and motion state classification in the later stages of the signal processing pipeline.

3.3 Block C: Signal Preprocessing

The acquired PPG and acceleration signals are filtered to remove baseline drift and high-frequency noise prior to subsequent processing. The acceleration data consist of the computed acceleration magnitude and the X-axis acceleration. The acceleration magnitude is used for motion metric generation, while the X-axis acceleration is used as the reference input for the adaptive noise cancellation stage. The X-axis acceleration was selected instead of using all three acceleration axes because it exhibited the strongest correlation with the motion artifacts present in the PPG signal and produced the highest heart-rate estimation accuracy during experimental evaluation as explained in detail in *Section 6*.

Signal conditioning is performed using cascaded second-order Direct Form II Transposed (DF2T) biquad high-pass and low-pass filters. A biquad filter is a second-order Infinite Impulse Response (IIR) filter capable of implementing precise frequency responses using only a small number of coefficients and state variables. Compared to FIR filters with equivalent frequency selectivity, biquad filters achieve similar filtering performance with significantly lower computational and memory requirements. The

Direct Form II Transposed structure was selected due to its reduced memory requirements and improved numerical robustness in finite-precision implementations. Together, the cascaded high-pass and low-pass filters form a band-pass response.

Both the PPG and acceleration signals are first processed using a high-pass filter with a cutoff frequency of 0.5 Hz to remove the DC component and low-frequency baseline variations caused by sensor placement, tissue deformation, gravitational acceleration, and slow body movements.

For the PPG signal, a low-pass filter with a cutoff frequency of 5 Hz is applied. Since the physiological heart-rate range is well below 300 BPM (5 Hz), the pulsatile cardiac component is preserved while higher-frequency noise and interference are attenuated. In contrast, the acceleration signals are filtered using a low-pass cutoff frequency of 10 Hz. This higher cutoff frequency preserves motion information associated with motion-induced artifacts in the PPG signal while suppressing higher-frequency sensor noise. Retaining a wider motion bandwidth improves the effectiveness of the adaptive noise cancellation stage by providing a more representative motion reference signal.

All filters were implemented as second-order DF2T biquad filters with a quality factor of ($Q = 0.7071$), corresponding to a second-order Butterworth response. This configuration provides a maximally flat passband response without introducing excessive resonance near the cutoff frequency. The filter coefficients were generated for a sampling frequency of 100 Hz and stored as fixed coefficients within the embedded implementation.

3.4 Block D: Motion Metric Computation & Motion-state Classification

The filtered acceleration magnitude ($a_{mag,bpf,i}$) is used to generate a motion metric that represents the overall intensity of motion within a fixed time window. In this block, the motion metric (M) is calculated as the root mean square (RMS) value of the filtered acceleration magnitude over a 100-sample window (N), according to

$$M = \sqrt{\frac{1}{N} \sum_{i=1}^N a_{mag,bpf,i}^2}$$

Since the system operates at a sampling rate of 100 Hz, this corresponds to a 1-second analysis window.

RMS is used because it provides a single positive measure of the overall motion intensity within the analysis window. Compared to a simple arithmetic average, RMS is insensitive to signal polarity and

emphasizes larger motion amplitudes due to the squaring operation. This makes it suitable for estimating the strength of user motion from the filtered acceleration magnitude.

The motion metric is updated every 50 samples, corresponding to an update interval of 0.5 seconds. To avoid unreliable motion estimates, the block only calculates the metric when at least 80% of the samples in the window are valid. If this condition is not satisfied, the motion metric is set to zero and the motion state is set to low.

	$M < 700$	$700 \leq M < 9000$	$M \geq 9000$
Motion States	LOW	MODERATE	HIGH

Table 1: Motion states of varying motion intensity

The calculated motion metric is then classified into three motion states as shown in Table 1. If ($M < 700$), the motion state is classified as low. If ($700 \leq M < 9000$), the motion state is classified as moderate. If ($M \geq 9000$), the motion state is classified as high. These thresholds were empirically selected through preliminary experiments conducted under stationary, moderate-motion, and severe-motion conditions to provide reliable separation between the different motion levels. This motion state is subsequently used to determine the operating mode of the NLMS adaptive noise cancellation block.

3.5 Block E: Motion-aware NLMS-based Adaptive Noise Cancellation

This block performs motion artifact reduction using an Adaptive Noise Cancellation (ANC) framework. ANC is a signal processing technique that removes undesired interference from a primary signal by utilizing a secondary reference signal that is correlated with the interference. In this work, the filtered PPG signal is used as the primary input, while the filtered X-axis acceleration signal is used as the reference input. The objective of the ANC stage is to estimate the motion-artifact component correlated with the acceleration reference signal and subtract it from the PPG signal, thereby producing a cleaner physiological waveform for subsequent heart-rate estimation.

Unlike conventional ANC implementations that continuously operate under a fixed adaptation strategy, the proposed system incorporates motion-state information obtained from Block D to

dynamically control the behavior of the adaptive filter. Depending on the detected motion level, the ANC stage can operate in bypass, adapt, or freeze mode. This motion-aware adaptation framework is a key feature of the proposed design and is discussed in detail later in this section.

The ANC filter is implemented as a Finite Impulse Response (FIR) filter. An FIR filter computes its output as a weighted sum of delayed input samples according to

$$\hat{v}(n) = \sum_{k=0}^{L-1} w_k(n) \cdot u(n - k)$$

where $u(n)$ is the filtered X-axis acceleration reference signal, $w_k(n)$ represents the adaptive filter coefficients, L is the filter length, and $\hat{v}(n)$ is the estimated motion artifact. FIR filters were selected because they are inherently stable and allow straightforward coefficient adaptation during real-time operation.

The estimated motion artifact is removed from the filtered PPG signal according to

$$e(n) = d(n) - \hat{v}(n)$$

where $d(n)$ represents the filtered PPG signal and $e(n)$ represents the motion-reduced PPG signal. The signal $e(n)$ is subsequently forwarded to the heart-rate estimation stage.

To adapt the FIR coefficients in real time, the Normalized Least Mean Squares (NLMS) algorithm is employed. NLMS is an adaptive filtering algorithm that iteratively updates the filter coefficients to minimize the output error signal. Compared to the conventional LMS algorithm, NLMS normalizes the adaptation step size using the reference signal power, improving convergence stability when the reference signal amplitude varies. The coefficient update equation is given by

$$w_k(n + 1) = w_k(n) + \frac{\mu}{\varepsilon + ||u(n)||^2} \cdot e(n) \cdot u(n - k)$$

where $||u(n)||^2$ is

$$||u(n)||^2 = \sum_{k=0}^{L-1} u^2(n - k)$$

μ is the adaptation step size, ϵ is a small positive constant used to prevent division by zero, and $\|u(n)\|^2$ denotes the reference signal power.

Unlike conventional ANC implementations that continuously adapt their coefficients, the proposed approach incorporates motion-state information from Block D to control the behavior of the adaptive filter. The motion state is classified as low, moderate, or high motion, and the ANC filter operates in one of three corresponding modes: bypass, adapt, and freeze. This motion-aware adaptation strategy forms the primary contribution of the proposed ANC framework.

During low-motion conditions, the filter operates in bypass mode. In this mode, adaptive filtering is disabled and the filtered PPG signal is passed directly to the output,

$$e(n) = d(n)$$

Since motion artifacts are expected to be minimal, bypassing the adaptive filter avoids unnecessary signal modification and preserves the original PPG waveform.

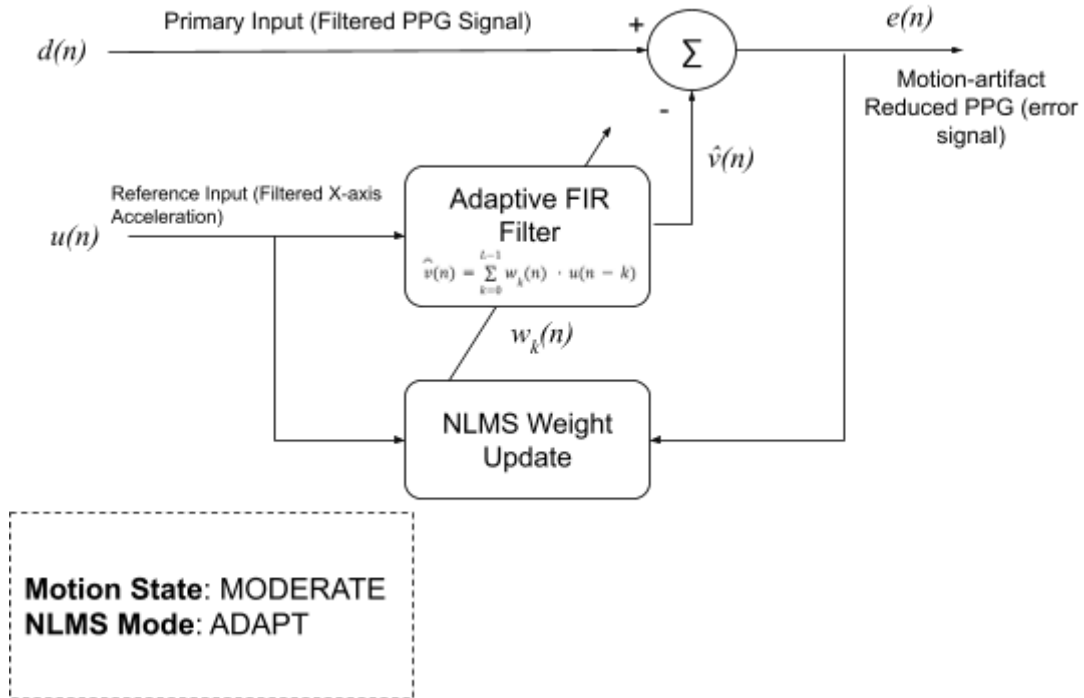


Figure 4.1: Signal Flow of ADAPT Mode ANC

During moderate-motion conditions, the filter operates in ADAPT mode. In this mode, the FIR filter estimates the motion artifact and the NLMS algorithm continuously updates the filter coefficients according to the adaptation equation. This allows the filter to learn the relationship between the acceleration reference signal and the motion artifact present in the PPG signal, enabling real-time artifact suppression.

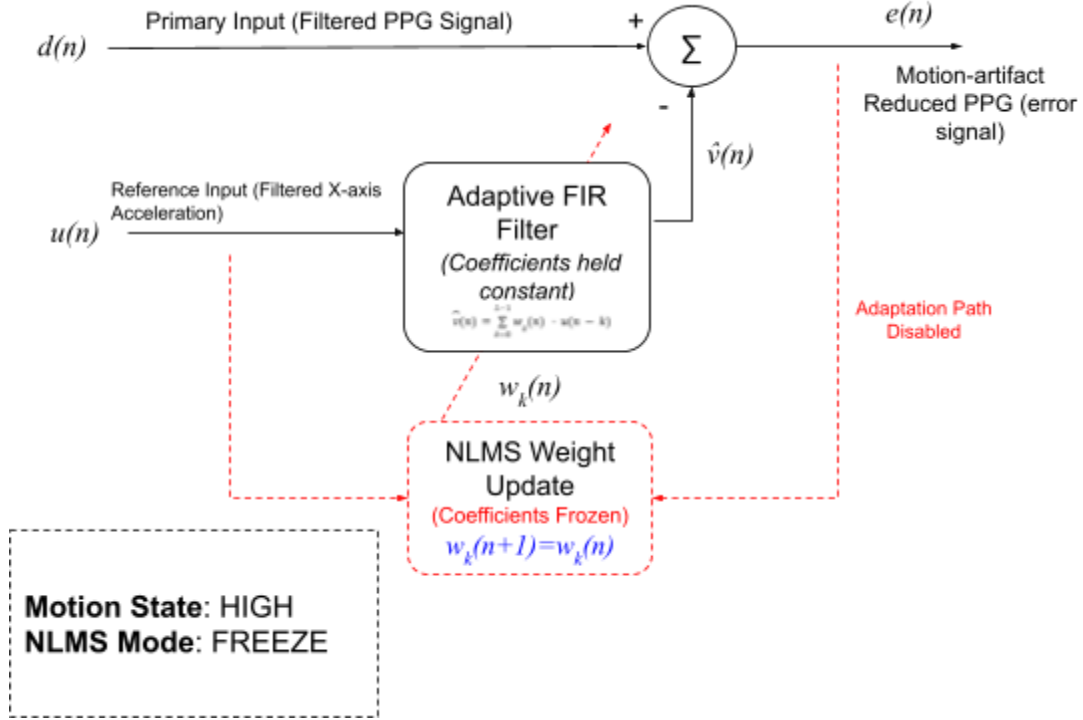


Figure 4.2: Signal Flow of FREEZE Mode ANC

During high-motion conditions, the filter operates in freeze mode. In this mode, the FIR filter continues to apply the previously learned coefficients to the acceleration reference signal, but coefficient adaptation is disabled. Consequently,

$$w_k(n + 1) = w_k(n)$$

and the coefficient vector remains unchanged. This prevents the adaptive filter from updating its coefficients during periods of excessive motion, where the relationship between the acceleration reference

signal and the PPG artifact may become unreliable. The previously learned coefficients are therefore retained until the motion level returns to a moderate state.

The resulting output of this block is a motion-reduced PPG signal that is subsequently used for heart-rate estimation.

3.6 Block F: FFT-based HR Estimation

Block F estimates heart rate from the motion-reduced PPG signal produced by Block E. In this project, the time-domain estimator was excluded, and the implemented heart-rate estimate is based on a frequency-domain Fast Fourier Transform (FFT) method.

An FFT-based estimator analyzes the frequency content of the PPG signal instead of detecting individual pulse peaks in the time domain. Since heart rate is a periodic physiological signal, it appears as a dominant frequency component in the PPG spectrum. This makes FFT-based estimation useful when the waveform shape is distorted by residual motion artifacts or noise.

The estimator operates on a 512-sample window of the cleaned PPG signal. With a sampling frequency of 100 Hz, this corresponds to an analysis window of 5.12 seconds.

3.6.1 Mean Removal

Before spectral analysis, the mean value of the input window is removed:

$$x_{dc}(n) = x(n) - \bar{x}$$

where $x(n)$ is the cleaned PPG signal and $\bar{x}$ is the mean value of the window. Mean removal eliminates the DC component so that the FFT is focused on the pulsatile part of the signal rather than the baseline offset.

3.6.2 RMS Signal Validation

The RMS value of the DC-removed signal is then calculated according to

$$RMS = \sqrt{\frac{1}{N} \sum_{n=0}^{N-1} x_{dc}^2(n)}$$

RMS provides a measure of the signal magnitude within the analysis window. If the RMS is below the minimum threshold, the window is rejected because the signal amplitude is too small for reliable frequency-domain heart-rate estimation.

3.6.3 Hann Windowing

After mean removal, a Hann window is applied before the FFT. The Hann window is defined as

$$h(n) = 0.5 - 0.5 \cos\left(\frac{2\pi n}{N-1}\right)$$

and the windowed signal is

$$x_w(n) = x_{dc}(n) \cdot h(n)$$

The Hann window reduces spectral leakage by tapering the beginning and end of the finite-length signal window. This improves the reliability of spectral peak detection.

3.6.4 FFT and Magnitude Spectrum

The FFT is used to efficiently compute the Discrete Fourier Transform:

$$X(k) = \sum_{n=0}^{N-1} x_w(n) \cdot e^{-j \frac{2\pi kn}{N}}$$

The magnitude spectrum is then calculated from the real and imaginary FFT components:

$$|X(k)| = \sqrt{\text{Re}\{X(k)\}^2 + \text{Im}\{X(k)\}^2}$$

This magnitude spectrum shows how strongly each frequency component is present in the cleaned PPG signal.

3.6.5 Multi-Frame Spectral Averaging

To reduce short-term spectral fluctuations, the FFT magnitude spectrum is averaged across frames. On the first valid frame, the averaged spectrum is initialized using the current magnitude spectrum. For later frames, the averaged spectrum is updated as

$$X_{avg}(k) = \frac{N_{avg}-1}{N_{avg}} \cdot X_{avg}(k) + \frac{1}{N_{avg}} \cdot |X(k)|$$

where ($N_{avg} = 3$). This averaging reduces random frame-to-frame variation and makes the selected heart-rate peak more stable.

3.6.6 Heart-Rate Search Region

The search range is limited to the expected heart-rate band of 40 BPM to 130 BPM. These limits are converted to frequency using

$$f_{min} = \frac{HR_{min}}{60}$$

$$f_{max} = \frac{HR_{max}}{60}$$

The corresponding FFT bin limits are

$$k_{min} = \left\lceil \frac{f_{min} \cdot N}{f_s} \right\rceil$$

$$k_{max} = \left\lceil \frac{f_{max} \cdot N}{f_s} \right\rceil$$

Only bins within this range are considered for heart-rate estimation, which reduces the chance of selecting irrelevant low-frequency drift or high-frequency noise.

3.6.7 Continuity-Based Peak Selection

Within the heart-rate search region, the algorithm searches for the dominant spectral peak. The score of each candidate bin is based on its magnitude, with an additional continuity bias from the previous valid FFT heart-rate estimate:

$$Score(k) = |X_{avg}(k)| \cdot \frac{1}{1+0.2|k-k_{prev}|}$$

where k_{prev} is the bin corresponding to the previous FFT heart-rate estimate. This favors peaks close to the previous estimate and reduces sudden jumps to unrelated spectral peaks.

Finally, the bin with the highest score is selected as the candidate heart-rate peak.

3.6.8 Harmonic and Local Peak Checks

After selecting the candidate peak, the estimator evaluates whether the selected bin has reasonable spectral structure.

First, harmonic consistency is checked by examining the magnitude near the second harmonic:

$$k_{harm} = 2k$$

A PPG waveform is not a perfect sinusoid, so harmonic components can appear at multiples of the fundamental heart-rate frequency. The presence of energy near the second harmonic increases confidence that the selected peak corresponds to a physiological rhythm.

Second, local peak validity is checked. The selected bin must be greater than its neighboring bins, meaning it should represent an actual local maximum rather than a flat region or random fluctuation.

3.6.9 Parabolic Peak Interpolation

The raw FFT frequency resolution is limited by the FFT length. To estimate the peak location more precisely, parabolic interpolation is applied using the selected bin and its two neighboring bins:

$$\delta = \frac{1}{2} \cdot \frac{M_{k-1} - M_{k+1}}{M_{k-1} - 2M_k + M_{k+1}}$$

where M_k is the magnitude at bin k . The interpolated peak bin is

$$K_{peak} = k + \delta$$

The peak frequency is then calculated as

$$f_{peak} = \frac{K_{peak} \cdot f_s}{N}$$

and converted to heart rate:

$$HR_{FFT} = 60 \cdot f_{peak}$$

If the resulting heart rate is outside the 40 BPM to 130 BPM range, the estimate is rejected.

3.6.10 Temporal Tracking

After calculating the FFT heart-rate estimate, temporal tracking is applied using the previous tracked FFT estimate:

$$HR_{tracked} = \alpha HR_{prev} + (1 - \alpha) HR_{FFT}$$

where ($\alpha=0.75$). This smoothing reduces short-term fluctuations between consecutive windows while still allowing the estimate to follow gradual changes in heart rate.

3.6.11 Confidence Estimation

After temporal tracking, the estimator calculates a confidence value for the detected spectral peak. The first component is spectral dominance, measured using the peak ratio:

$$R = \frac{M_{peak}}{mean(M_k)}$$

where M_{peak} is the selected peak magnitude and $mean(M_k)$ is the average magnitude within the heart-rate search band. If this ratio is below the minimum threshold, the estimate is rejected.

The ratio score is calculated as

$$C_{ratio} = \frac{R - R_{min}}{R_{span}}$$

and clamped to the range $([0,1])$.

The final confidence value combines spectral dominance, harmonic consistency, and local peak validity:

$$C_{FFT} = 0.60 \cdot C_{ratio} + 0.25 \cdot C_{harmonic} + 0.15 \cdot C_{local}$$

where $C_{harmonic}$ is the harmonic consistency score and C_{local} is set to 1 when the selected bin is a valid local peak and 0 otherwise.

The output of this block is the tracked FFT heart-rate estimate, the FFT confidence value, the selected peak bin, the peak frequency, and the peak ratio.

3.7 Block G: Signal Quality Index

Block G evaluates the reliability of the FFT-based heart-rate estimate generated in Block F. The purpose of this block is to prevent unreliable heart-rate estimates from being forwarded to the final output stage.

The quality evaluation is primarily based on the confidence score and peak ratio produced by the FFT estimator. The confidence score reflects the overall quality of the detected spectral peak, while the peak ratio indicates how dominant the selected heart-rate peak is within the frequency spectrum. Estimates with low confidence or weak spectral dominance are classified as unreliable.

A motion quality score is also calculated from the motion metric generated in Block D. This score provides an indication of the current motion level and is normalized between 0 and 1, where larger values correspond to lower motion levels.

Based on these quality measures, the block determines whether the FFT-based heart-rate estimate is valid. The resulting quality indicators and validity flags are then passed to the final decision stage for heart-rate output generation.

3.8 Block H: Decision Logic

Block H generates the final heart-rate output of the system. If the FFT-based heart-rate estimate is classified as valid, the estimate and its confidence value are accepted as the current output. Otherwise, the system enters a hold mode and retains the previously accepted heart-rate value. This prevents unreliable estimates from causing sudden fluctuations in the reported heart rate and improves output stability.

4. Hardware

4.1 Circuit & Electrical Wiring

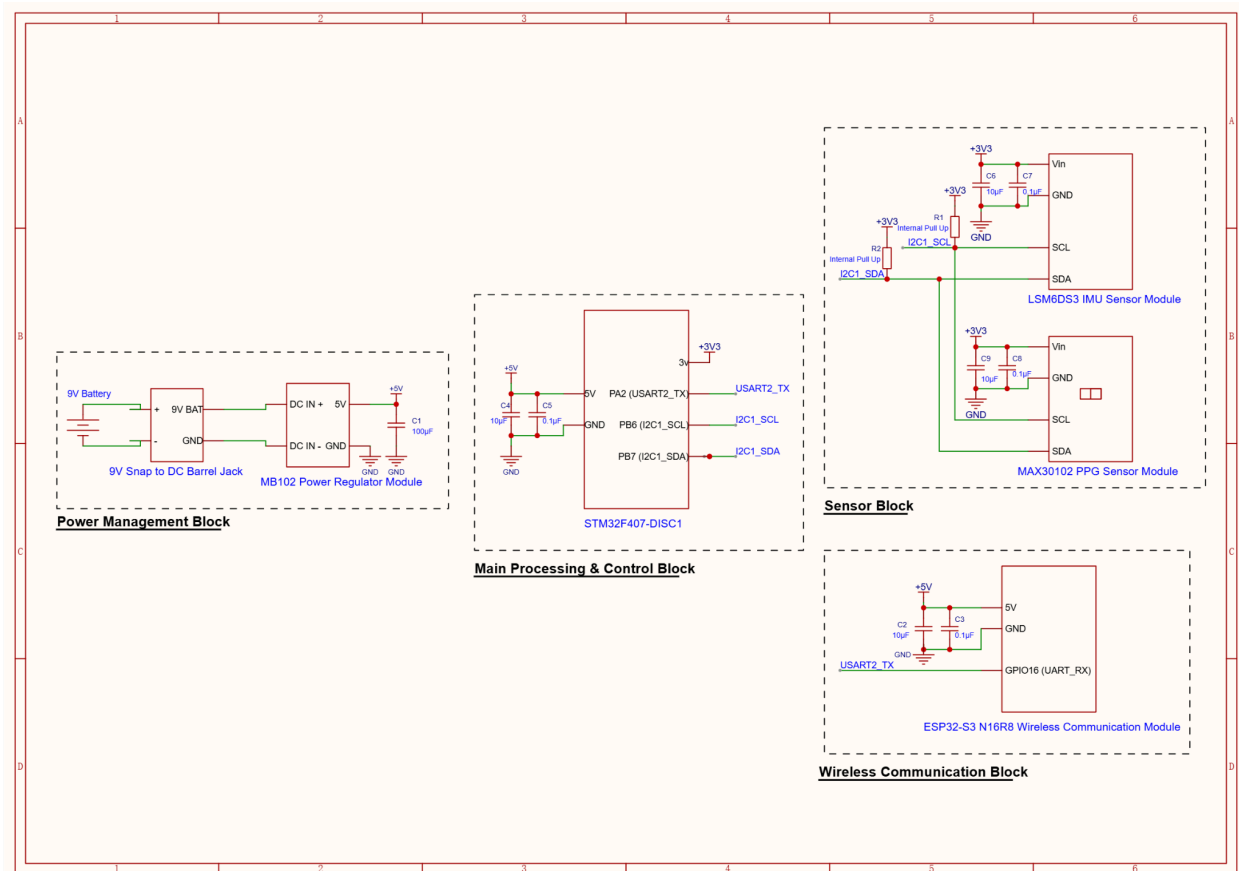


Figure 5: Project's Hardware

Figure 5 is a schematic diagram illustrating the hardware components used and their electrical wirings. The functional blocks consist of power management block, main processing and control block, sensor block, and wireless communication block.

4.1.1 Power Management Block

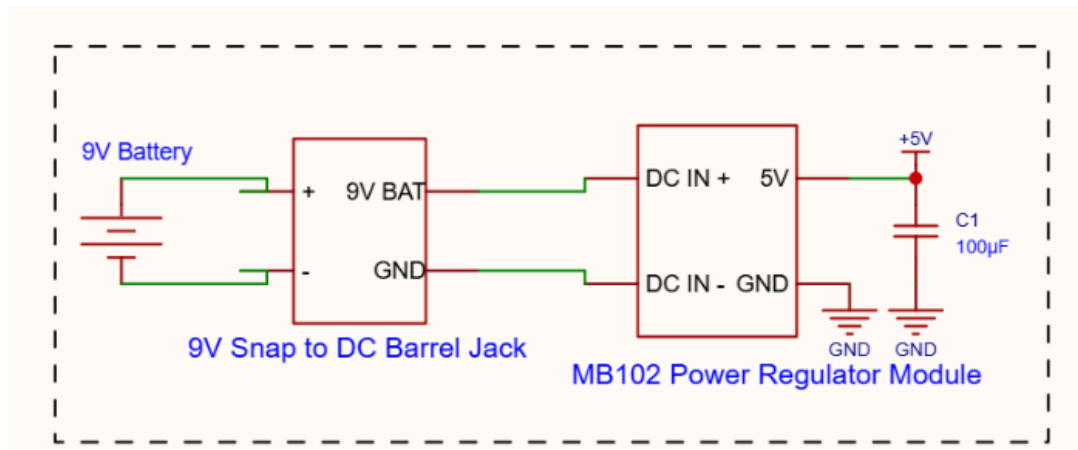


Figure 5.1: Power Management Block

As shown in *Figure 5.1*, the system is powered by a 9 V battery connected to an MB102 power regulator module through a DC barrel jack. The regulator converts the input voltage to a stable 5 V supply, which powers both the STM32F407 microcontroller and the ESP32 wireless communication module through their respective 5 V input pins. To improve supply stability and reduce voltage fluctuations caused by transient load conditions, a 100 μ F bulk capacitor is placed at the regulator output. This power architecture provides a regulated voltage source for reliable operation of the embedded system.

4.1.2 Main Processing & Control Block

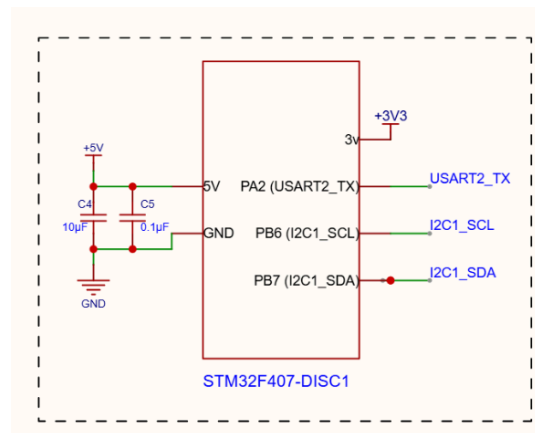


Figure 5.2: Main Processing & Control Block

The STM32F407 microcontroller serves as the main processing unit of the system and is responsible for real-time data acquisition and digital signal processing. The microcontroller provides a regulated 3.3 V supply to power the MAX30102 PPG sensor and the LSM6DS3 IMU sensor. Communication between the STM32F407 and the sensor modules is established through the I²C interface, while UART communication is used to transmit processed data to the ESP32 wireless communication module. To ensure stable operation and suppress supply noise, decoupling and bypass capacitors are placed between the power and ground rails near the microcontroller power pins.

4.1.3 Sensor Block

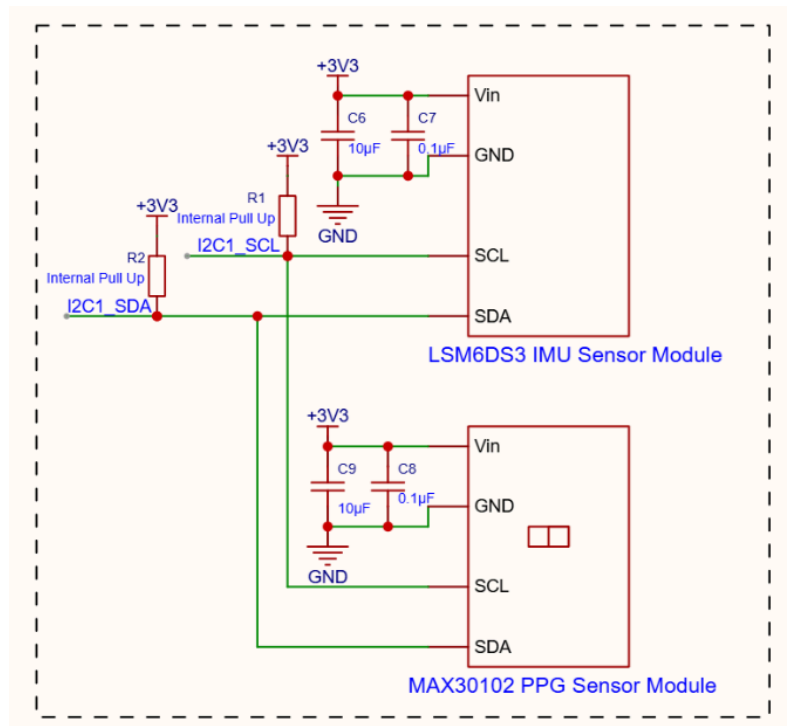


Figure 5.3: Sensor Block

The LSM6DS3 IMU sensor integrates a three-axis accelerometer and a three-axis gyroscope, although only accelerometer data are utilized in this work. The MAX30102 PPG sensor incorporates integrated red (660 nm) and infrared (880 nm) LEDs together with a photodetector for optical pulse measurement. Both sensors communicate with the MCU through the I²C interface. The I²C bus is

maintained in the logic-high state using the MCU's internal pull-up resistors. To ensure stable sensor operation and suppress supply noise, bypass and decoupling capacitors are connected between the supply rail and ground near each sensor module.

4.1.4 Wireless Communication Block

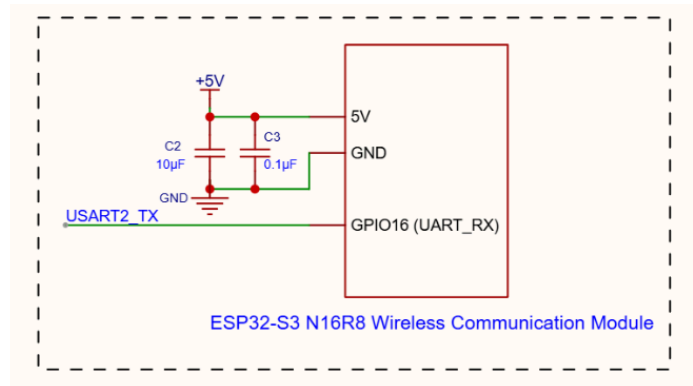


Figure 5.4: Wireless Communication Block

An ESP32-S3 N16R8 module is used to provide wireless communication between the embedded system and a host computer. Heart rate estimates, confidence scores, and other evaluation metrics are transmitted from the STM32F407 microcontroller to the ESP32 through a UART interface. The ESP32 operates as a Wi-Fi access point and hosts a TCP server, enabling real-time data logging, visualization, and monitoring on a connected computer. The module is powered by the regulated 5 V supply provided by the power subsystem. To ensure stable operation and suppress supply noise, bypass and decoupling capacitors are connected between the power rail and ground near the module.

4.2 Hardware Implementation

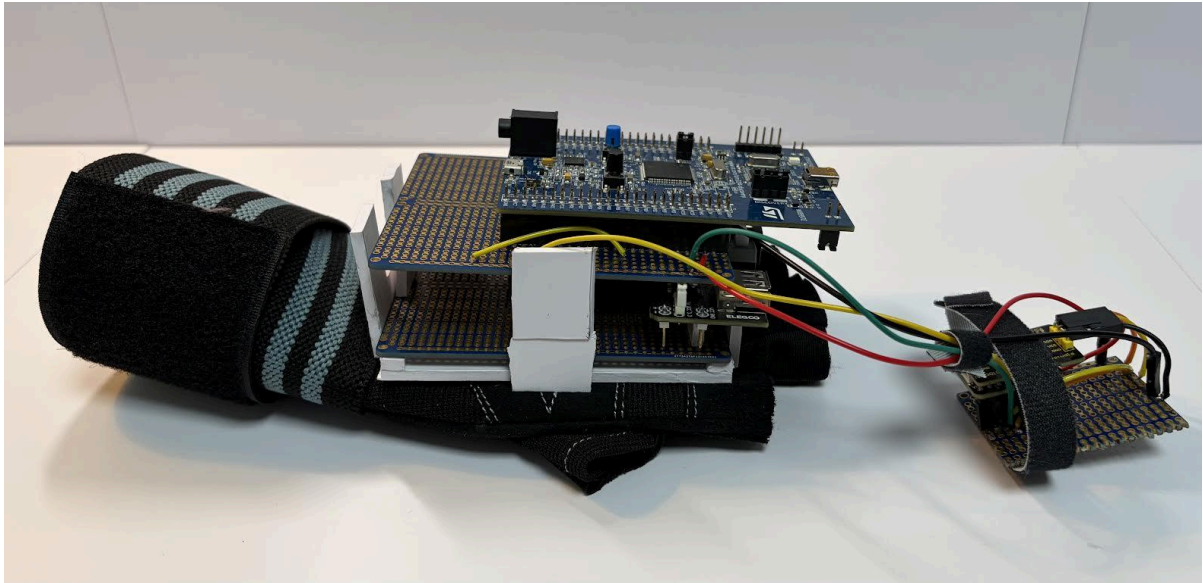


Figure 6.1: Project's Implemented Hardware

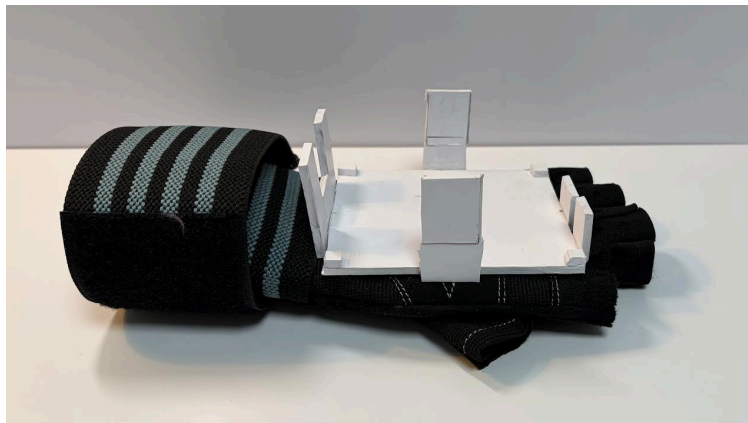


Figure 6.2: Glove-based Support Structure

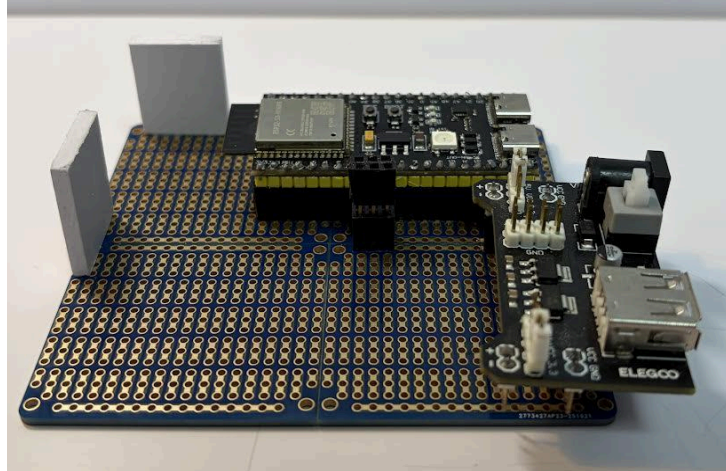


Figure 6.3: Platform's First Layer

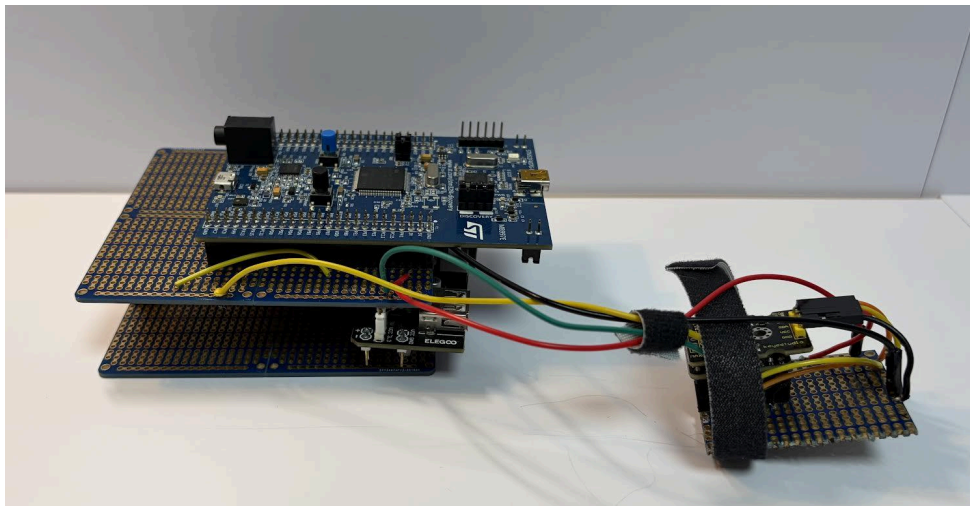


Figure 6.4: Platform's Two Layers stacked

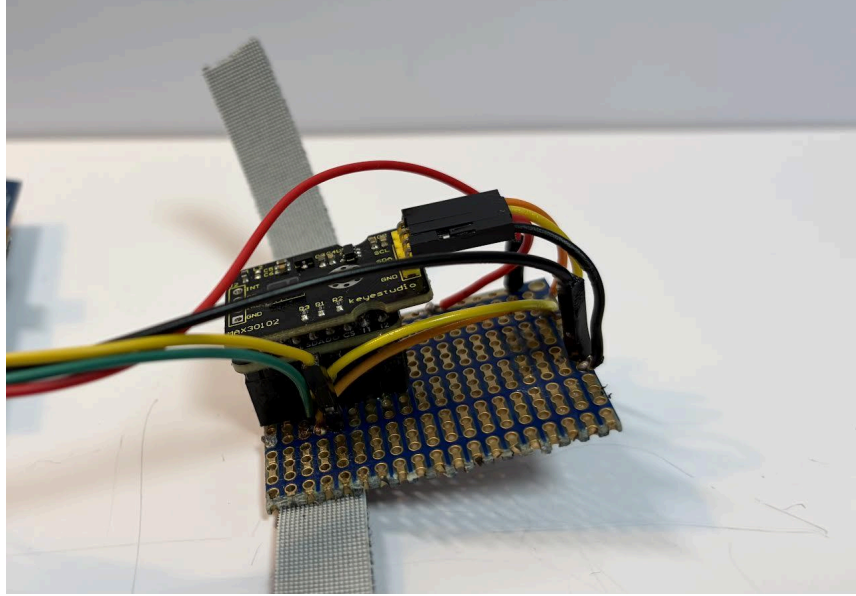


Figure 6.5: IMU and PPG Sensors

To facilitate wearable integration and hardware assembly, a custom mounting platform was constructed and attached to a glove-based support structure as shown in *Figure 6.2*. The platform provides a stable mechanical interface for mounting the electronic components, including the microcontroller, sensor modules, power supply, and communication module. The design allows individual hardware modules to be securely attached while maintaining accessibility for wiring, testing, and future modifications. The glove serves as the wearable base, whereas the elevated mounting structure acts as a modular framework that supports component placement and system integration during prototyping and experimental evaluation.

The wearable mounting structure consists of a two-layered platform that can be assembled and disassembled to facilitate system integration, testing, and maintenance. As shown in *Figure 6.3*, the first layer houses the wireless communication module and the power regulation circuitry. The second layer contains the STM32F407 microcontroller, positioned to provide convenient access for programming, debugging, and data acquisition during development.

The sensor subsystem is implemented as a separate module connected to the main processing platform. As shown in *Figure 6.5*, the LSM6DS3 IMU sensor is mounted directly beneath the MAX30102 PPG sensor to maximize the correlation between measured motion and motion artifacts present in the PPG signal. The sensor module is attached to the user's finger using a Velcro fastening mechanism, ensuring secure contact between the sensor assembly and the measurement site while minimizing relative motion during operation.

5. Experimental Setup

5.1 Finger placement

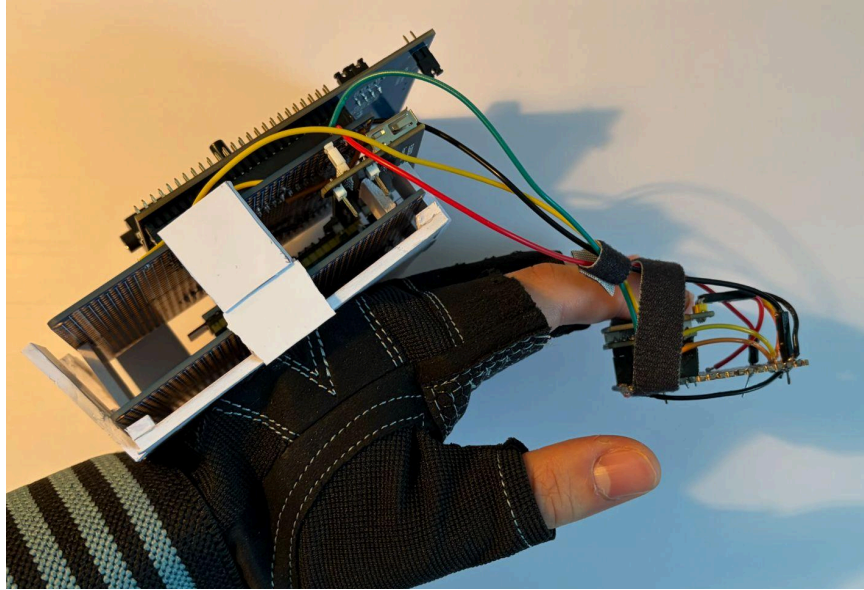


Figure 7: Finger Placement on the Sensors

The proposed heart-rate monitoring system was evaluated using fingertip photoplethysmography (PPG) measurements. During all experiments, the participant inserted the index finger into the sensor holder such that the fingertip was placed directly above the MAX30102 PPG sensor. To ensure consistent measurements, the finger was secured using a Velcro strap, minimizing relative motion between the skin surface and the sensor module. As shown in *Figure 7*, the PPG sensor and IMU module were mounted together on a dedicated fingertip platform, allowing the acceleration measurements to capture motion that directly affected the acquired PPG signal. The participant maintained the same finger placement throughout all experimental trials to ensure repeatability and consistency across different motion conditions.

5.2 External Heart Rate Reference Measurement Device

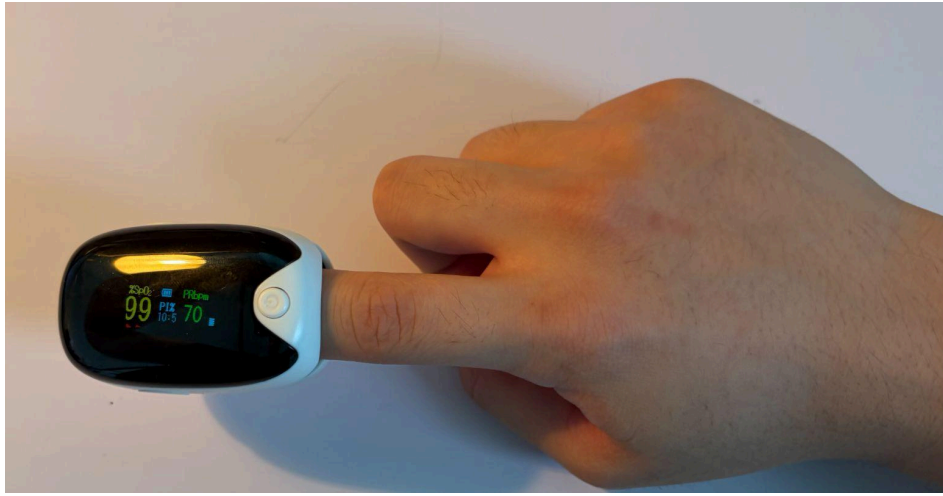


Figure 8: External Heart Rate Reference Estimator Device

To evaluate the accuracy of the proposed system, heart-rate measurements were compared against an external reference device. A commercially available fingertip pulse oximeter was used to provide reference heart-rate values during all experiments, as shown in *Figure 8*. The pulse oximeter was attached to the index finger of the participant's opposite hand to avoid interference with the PPG measurements acquired by the proposed system.

Reference heart-rate values were manually recorded at one-second intervals throughout each experiment and subsequently synchronized with the corresponding estimates produced by the proposed framework. The recorded reference values were then used to calculate performance metrics, including the mean absolute error (MAE), root mean square error (RMSE), and maximum estimation error. Using an independent optical heart-rate measurement device provided a practical benchmark for evaluating the accuracy and robustness of the proposed system under different motion conditions.

5.3 Performance Evaluation Metric

The performance of the proposed heart-rate monitoring system was evaluated using the mean absolute error (MAE), root mean square error (RMSE), maximum absolute error, heart-rate standard deviation, mean confidence score, and hold-mode percentage. These metrics were calculated by comparing the estimated heart rate produced by the proposed framework with the reference heart rate obtained from the external monitoring device.

The estimation error for each analysis window was defined as

$$e_i = |HR_{est,i} - HR_{ref,i}|$$

where $HR_{est,i}$ is the estimated heart rate and $HR_{ref,i}$ is the reference heart rate obtained from the external monitoring device.

The MAE was calculated as

$$MAE = \frac{1}{N} \sum_{i=1}^N e_i$$

while the RMSE was calculated as

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (HR_{est,i} - HR_{ref,i})^2}$$

where N is the number of analysis windows.

To evaluate the stability of the reported heart-rate output, the standard deviation of the estimated heart rate was also calculated. In addition, the mean confidence score generated by the signal quality assessment stage was recorded to assess the reliability of the frequency-domain heart-rate estimate. Finally, the hold-mode percentage was computed as the proportion of analysis windows in which the system retained the previous heart-rate estimate instead of accepting a new estimate, providing a measure of how frequently unreliable estimates were rejected.

6. Result & Evaluation

6.1 Reference Signal Selection via Correlation Analysis

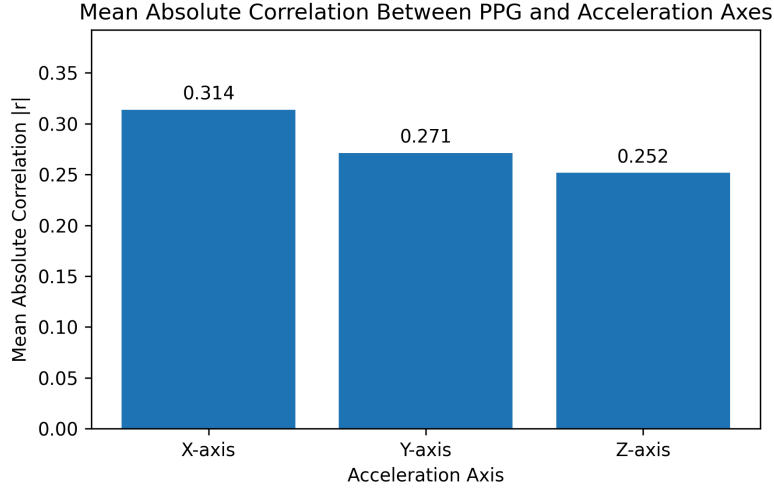


Figure 9: Mean Absolute Correlation Between PPG and Acceleration Axes

Prior to evaluating heart-rate estimation performance, a correlation analysis was conducted as an initial reference-signal selection experiment for adaptive noise cancellation. During a 1 Hz hand-swinging motion experiment, the filtered PPG signal was compared independently with the filtered X-, Y-, and Z-axis acceleration signals. The Pearson correlation coefficient was calculated for each valid analysis window according to

$$r = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^N (x_i - \bar{x})^2} \cdot \sqrt{\sum_{i=1}^N (y_i - \bar{y})^2}}$$

where x_i and y_i represent the filtered PPG and acceleration samples, respectively.

The absolute correlation value ($|r|$) was used because both positive and negative correlations indicate a relationship between the acceleration signal and motion-related variations in the PPG signal. A total of 215 valid analysis windows were evaluated. The X-axis acceleration exhibited the highest mean absolute correlation with the PPG signal, with ($|r| = 0.314$), compared with ($|r| = 0.271$) for the Y-axis and ($|r| = 0.252$) for the Z-axis. The X-axis was also identified as the most highly correlated axis in 102 out of 215 windows.

Based on these results, the X-axis acceleration was selected as the primary candidate reference signal for the NLMS adaptive noise cancellation stage. However, this correlation analysis was used only

as a heuristic reference-selection method and not as proof of optimal adaptive filtering performance. Although a stronger correlation may indicate a greater relationship between the acceleration signal and motion-induced PPG variations, adaptive noise cancellation performance depends not only on correlation magnitude but also on the temporal relationship between the reference signal and the motion artifact. Therefore, the X-axis NLMS configuration was subsequently evaluated against a 3-axis NLMS configuration to determine its actual effectiveness for heart-rate estimation under motion conditions.

6.2 Adaptive Noise Cancellation Performance Evaluation

To validate the correlation-based selection of the X-axis acceleration signal as the reference input for adaptive noise cancellation, the proposed X-axis NLMS configuration was compared against both a 3-axis NLMS configuration and a baseline configuration without adaptive noise cancellation. Heart-rate estimation performance was evaluated under controlled hand-swinging motion at frequencies of 1 Hz, 2 Hz, and 3 Hz. The mean absolute error (MAE) and root mean square error (RMSE) between the estimated heart rate and the reference heart rate were used as the primary performance metrics.

6.2.1 1 Hz Motion

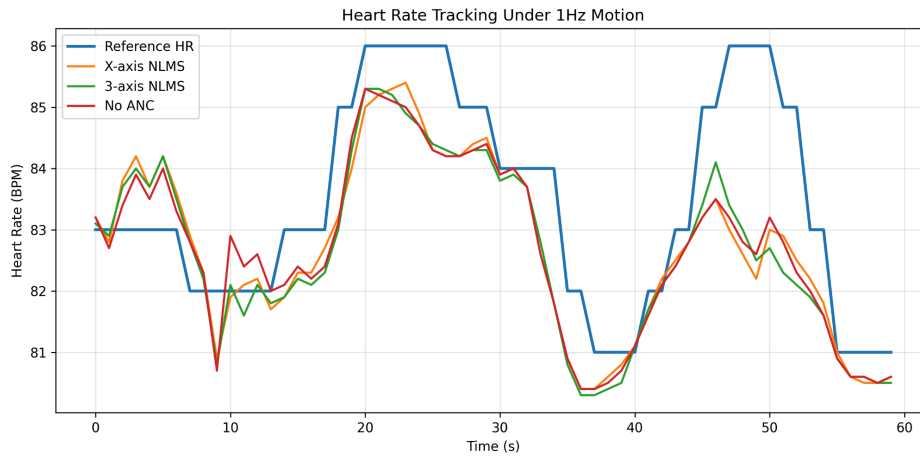


Figure 10.1: Time-domain comparison of the reference heart rate and heart-rate estimates obtained using the baseline, X-axis NLMS, and 3-axis NLMS configurations under the 1 Hz motion condition.

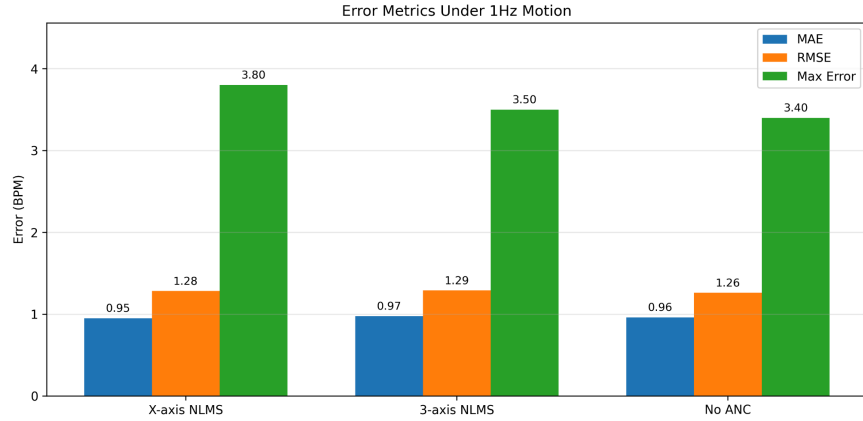


Figure 10.2: Error Metrics MAE, RMSE, and Maximum Error Under 1 Hz Motion

Under the 1 Hz motion condition, all three configurations exhibited nearly identical heart-rate estimation performance. As shown in *Figure 10.2*, the X-axis NLMS, 3-axis NLMS, and baseline (no adaptive noise cancellation) configurations achieved MAEs of 0.95 BPM, 0.97 BPM, and 0.96 BPM, respectively, corresponding to a maximum difference of only 0.02 BPM. Similarly, the RMSE values were 1.28 BPM, 1.29 BPM, and 1.26 BPM for the X-axis NLMS, 3-axis NLMS, and baseline configurations, respectively.

The negligible differences in both MAE and RMSE indicate that motion artifacts were minimal under the 1 Hz motion condition and therefore had limited impact on heart-rate estimation accuracy. These results suggest that the preprocessing pipeline alone was sufficient to suppress low-frequency motion-induced disturbances, resulting in comparable performance regardless of whether adaptive noise cancellation was applied.

6.2.2 2 Hz Motion

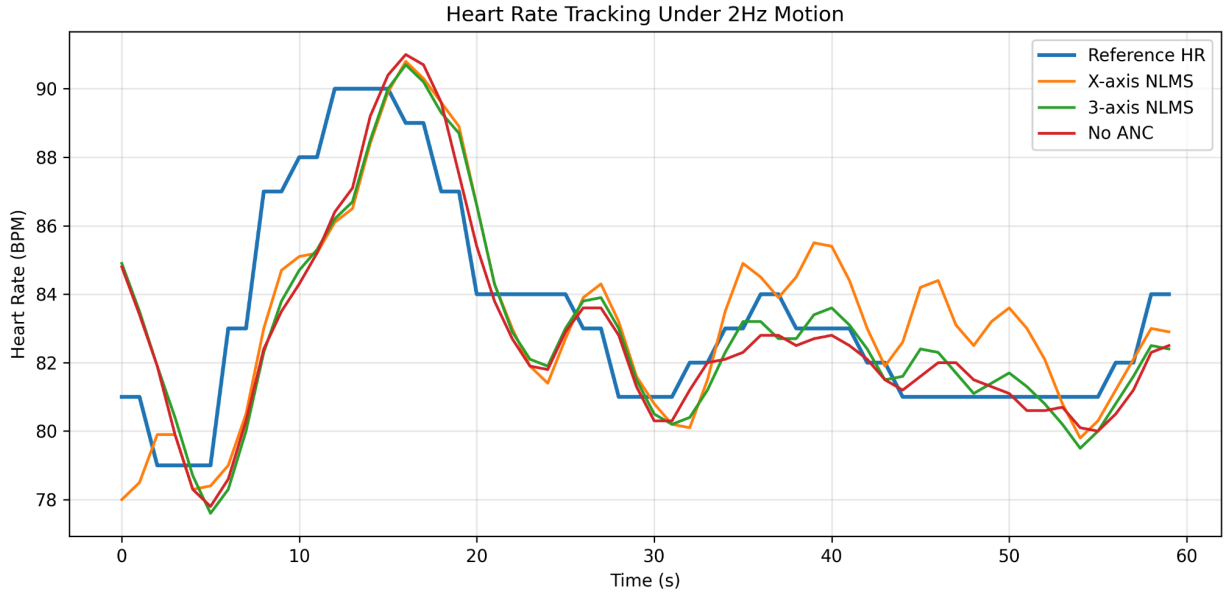


Figure 11.1: Time-domain comparison of the reference heart rate and heart-rate estimates obtained using the baseline, X-axis NLMS, and 3-axis NLMS configurations under the 2 Hz motion condition.

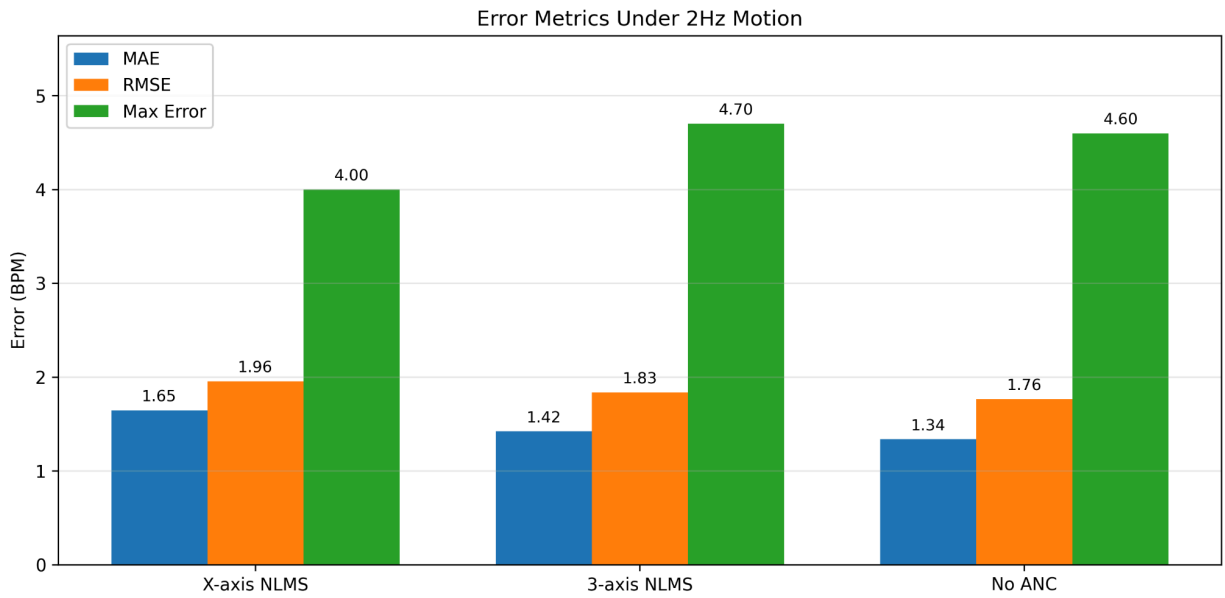
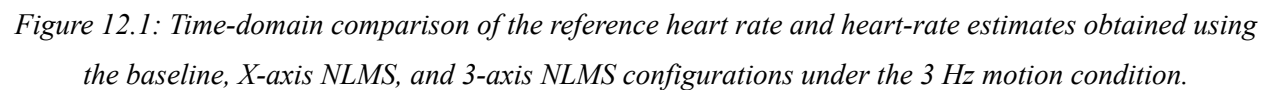


Figure 11.2: Error Metrics MAE, RMSE, and Maximum Error Under 2 Hz Motion

Under the 2 Hz motion condition, the baseline configuration achieved the lowest MAE of 1.34 BPM, followed by the 3-axis NLMS configuration with an MAE of 1.42 BPM and the X-axis NLMS

6.2.3 3 Hz Motion



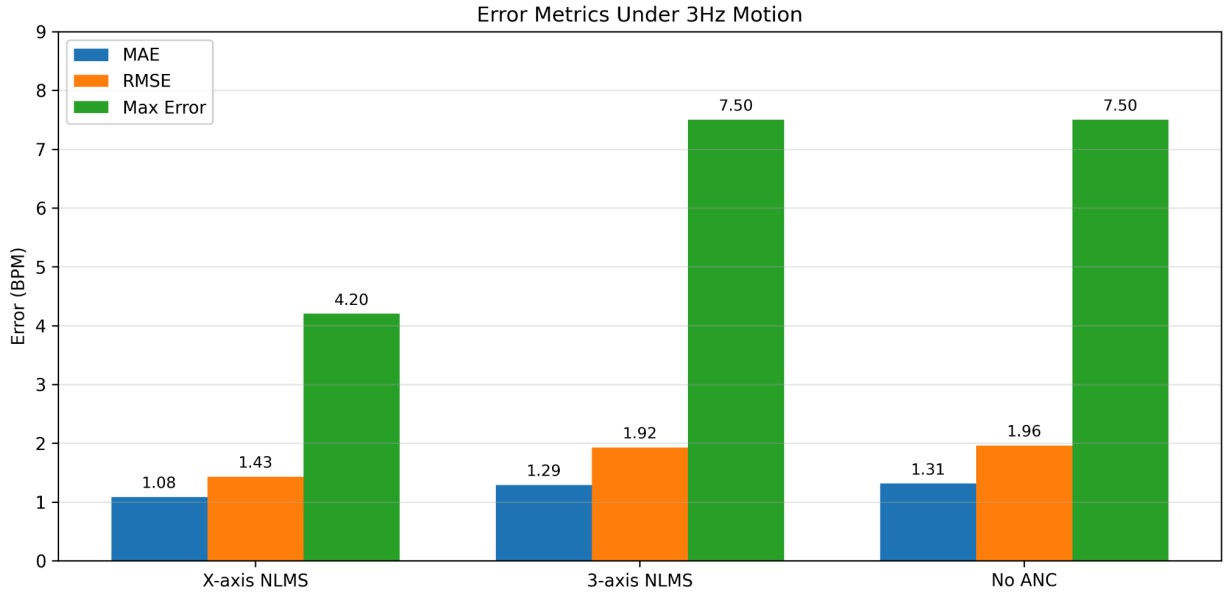


Figure 12.2: Error Metrics MAE, RMSE, and Maximum Error Under 3 Hz Motion

The largest performance differences were observed under the 3 Hz motion condition. As shown in *Figure 12.2*, the X-axis NLMS configuration achieved the lowest MAE of 1.08 BPM, outperforming both the 3-axis NLMS configuration (1.29 BPM) and the baseline configuration (1.31 BPM). A similar trend was observed in the RMSE results, where the X-axis configuration achieved the lowest RMSE of 1.43 BPM compared to 1.92 BPM and 1.96 BPM for the 3-axis and baseline configurations, respectively.

Unlike the 1 Hz and 2 Hz motion conditions, where all configurations produced comparable results, the benefits of adaptive noise cancellation became more apparent at higher motion intensities. This suggests that the severity of motion artifacts influences the effectiveness of the NLMS filter. When motion levels are low, the preprocessing and heart-rate estimation pipeline are generally sufficient to maintain accurate heart-rate estimates, leaving little opportunity for adaptive filtering to improve performance. In such cases, adaptive noise cancellation may even introduce slight performance degradation by attenuating useful cardiac components alongside motion-related interference. However, as motion intensity increases and motion artifacts become a more dominant source of signal corruption, adaptive filtering becomes increasingly beneficial, resulting in improved estimation accuracy.

Although the correlation analysis was originally introduced solely as a reference-signal selection method, the experimental results further validate the use of the X-axis acceleration signal as the primary NLMS reference input. Across all motion conditions, the X-axis configuration consistently achieved performance comparable to or better than the 3-axis configuration, while producing the best overall results under the most challenging 3 Hz motion condition. Furthermore, a single-axis reference reduces

computational complexity, memory requirements, and adaptive filter dimensionality, making it more suitable for real-time embedded implementation. Therefore, the X-axis acceleration signal was selected as the reference input for the final implementation of the adaptive noise cancellation stage.

6.3 Final HR Estimation Performance

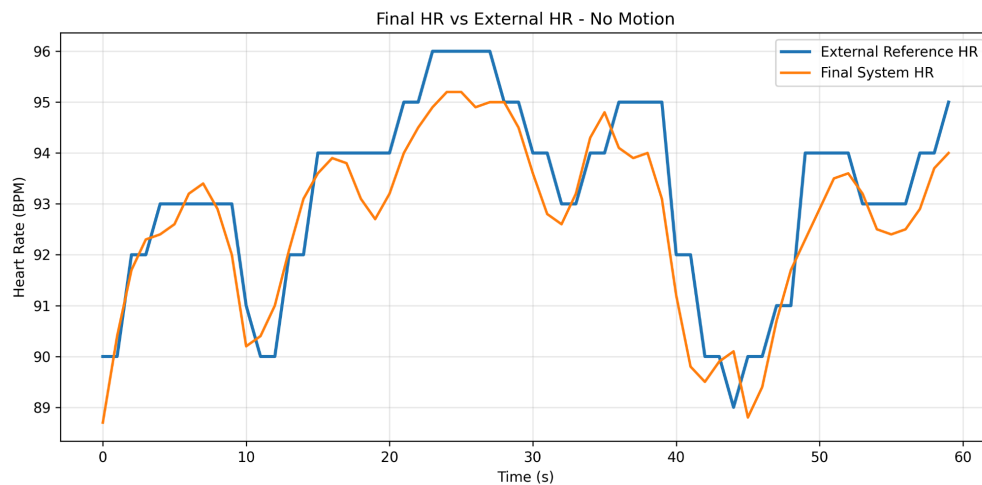


Figure 13.1: Time-domain comparison of the final heart-rate output and the external reference heart rate under the no motion condition.

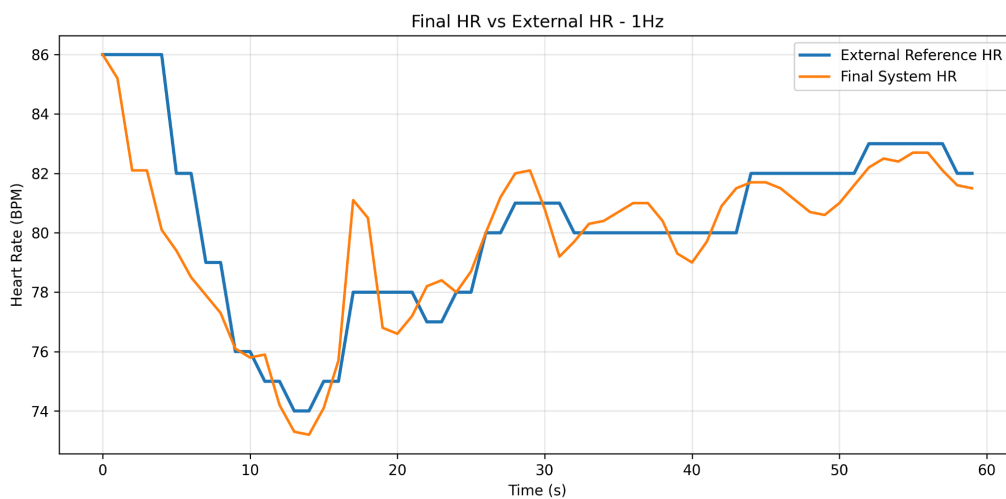


Figure 13.2: Time-domain comparison of the final heart-rate output and the external reference heart rate under the 1 Hz motion condition.

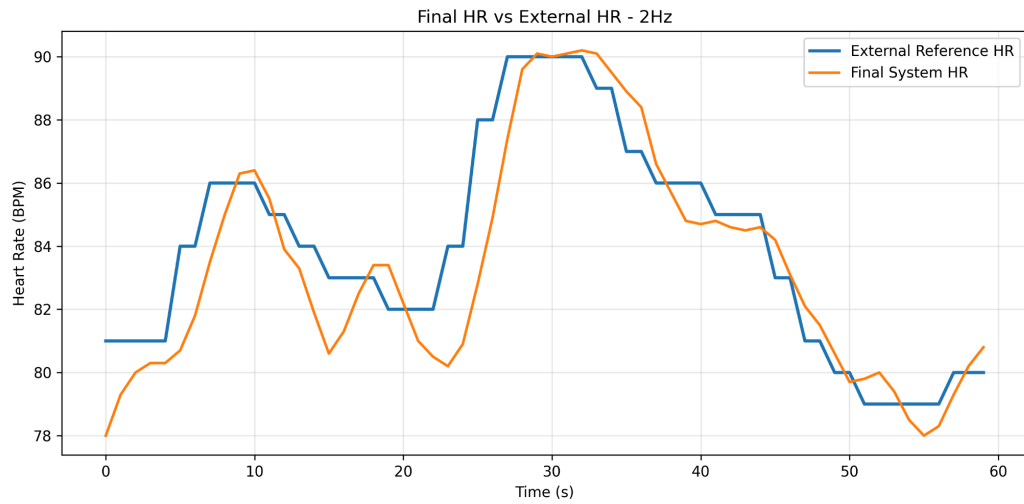


Figure 13.3: Time-domain comparison of the final heart-rate output and the external reference heart rate under the 2 Hz motion condition.

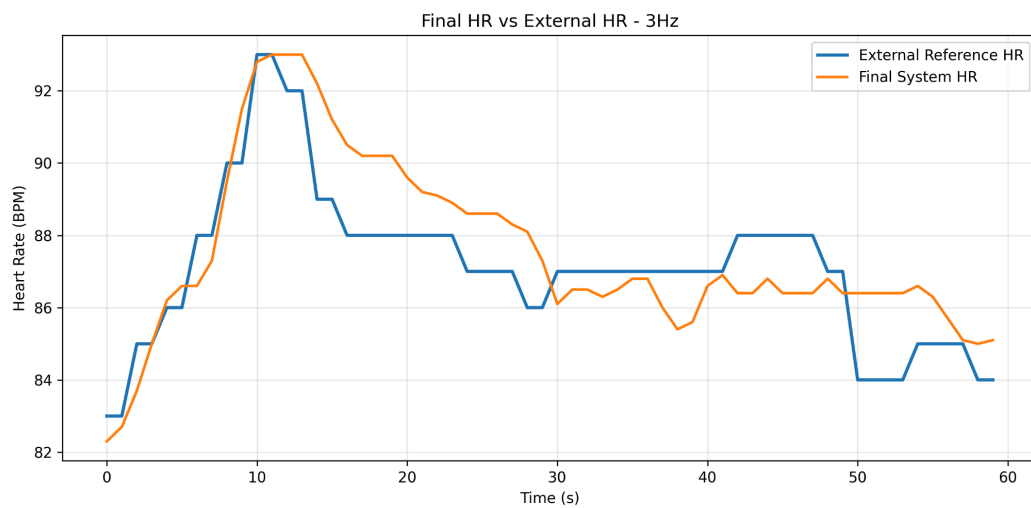


Figure 13.4: Time-domain comparison of the final heart-rate output and the external reference heart rate under the 3 Hz motion condition.

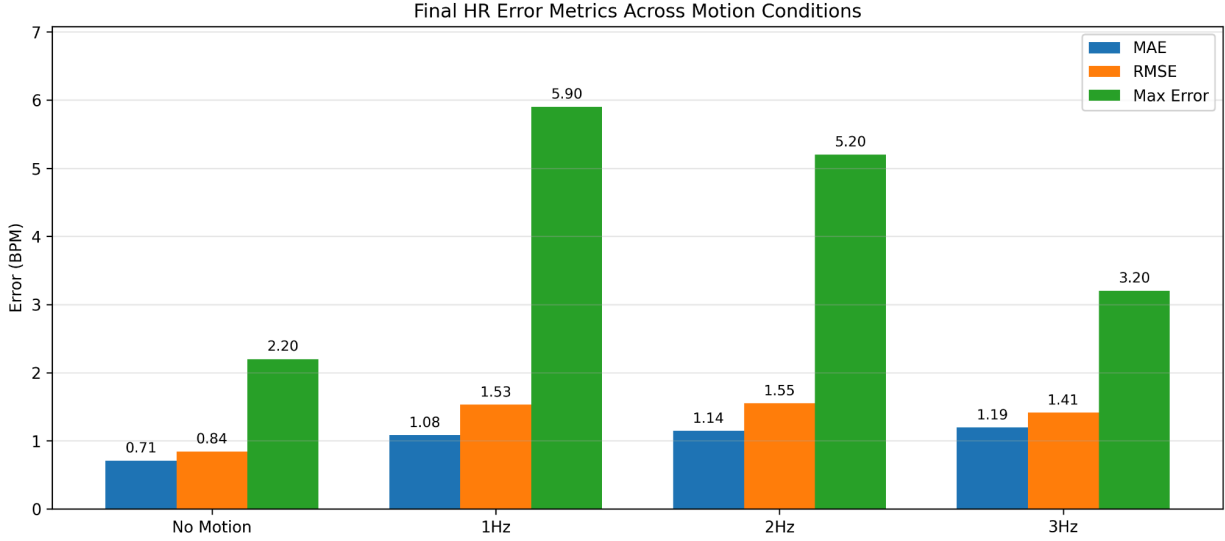


Figure 14: Error Metrics between Final HR and Reference HR across Motion Conditions

To evaluate the robustness of the proposed heart-rate estimation system against motion artifacts, experiments were conducted under four motion conditions: no motion, hand-swinging motion at 1 Hz, hand-swinging motion at 2 Hz, and hand-swinging motion at 3 Hz. These conditions introduced progressively greater levels of motion-induced interference into the acquired PPG signal. For each condition, 60 analysis samples were collected and compared against an external heart-rate reference device. The no-motion condition served as the baseline measurement, while the three hand-swinging experiments represented increasing levels of motion contamination.

Figures 13.1–13.4 present the time-domain comparison between the final heart-rate output and the external reference heart rate for each motion condition, while *Figure 14* summarizes the corresponding error metrics. Across all experiments, the estimated heart rate closely tracked the reference heart rate, indicating that the proposed framework maintained accurate heart-rate estimation under varying motion intensities.

The highest accuracy was achieved under the no-motion condition, with an MAE of 0.71 BPM and an RMSE of 0.84 BPM. The corresponding time-domain plot shows excellent agreement between the estimated and reference heart rates throughout the measurement period. The mean confidence score was also highest at 0.847, reflecting the strong spectral quality of the PPG signal in the absence of motion artifacts.

As motion intensity increased, estimation accuracy decreased slightly. The MAE increased from 1.08 BPM at 1 Hz to 1.14 BPM at 2 Hz and 1.19 BPM at 3 Hz, while the RMSE remained within a narrow range of 1.41–1.55 BPM. Despite the increasing motion contamination, the estimated heart rate

continued to follow the reference trend closely, with only small deviations observed in the time-domain plots.

A gradual reduction in the mean confidence score was observed with increasing motion intensity, decreasing from 0.605 at 1 Hz to 0.574 at 2 Hz and 0.473 at 3 Hz. This indicates that motion artifacts progressively degraded the spectral quality of the PPG signal. However, the corresponding increase in estimation error was relatively small, with the MAE increasing by only 0.11 BPM between the 1 Hz and 3 Hz conditions. This suggests that the proposed framework remained effective in maintaining estimation accuracy despite reduced signal quality.

The error-metric comparison shown in *Figure 14* further demonstrates the consistency of the proposed framework. The maximum error was 5.9 BPM at 1 Hz, 5.2 BPM at 2 Hz, and 3.2 BPM at 3 Hz. In addition, the heart-rate standard deviation remained within a relatively narrow range of 2.44–3.54 BPM across all motion conditions, indicating stable output behaviour despite the presence of motion artifacts.

Overall, the results demonstrate that the proposed motion-aware heart-rate estimation framework maintained accurate and stable heart-rate tracking across all tested motion conditions. The close agreement between the estimated and reference heart-rate traces, together with the low MAE and RMSE values, indicates that the framework effectively mitigates the impact of motion-induced interference on heart-rate estimation.

6.4 Effectiveness of Signal Quality Assessment & Hold-Mode Processing

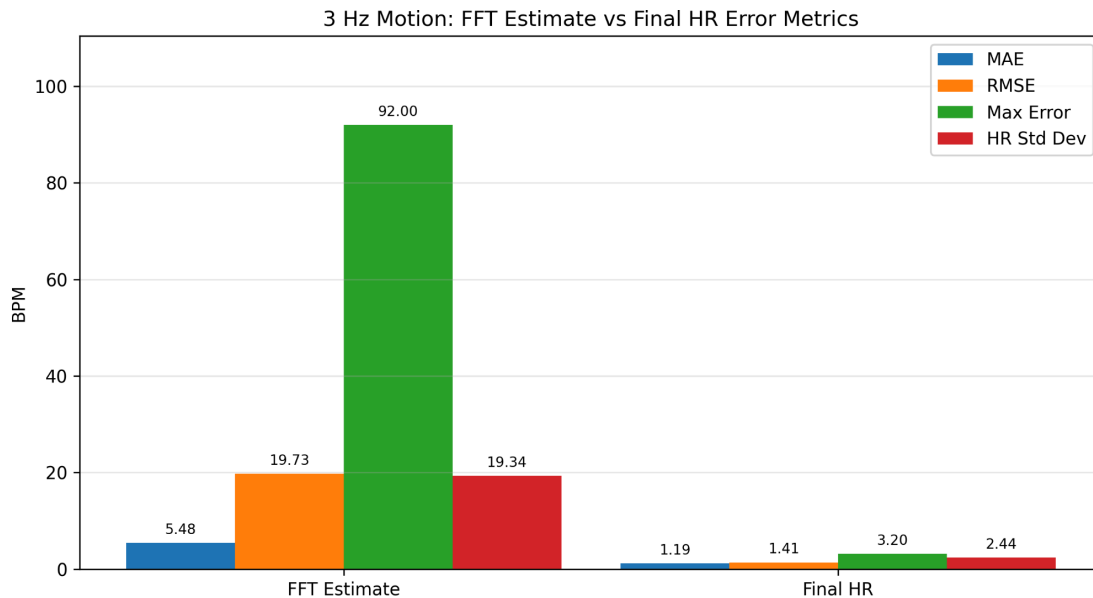


Figure 15: Comparison of Error Metrics between raw FFT estimate HR and Final HR under 3 Hz Motion

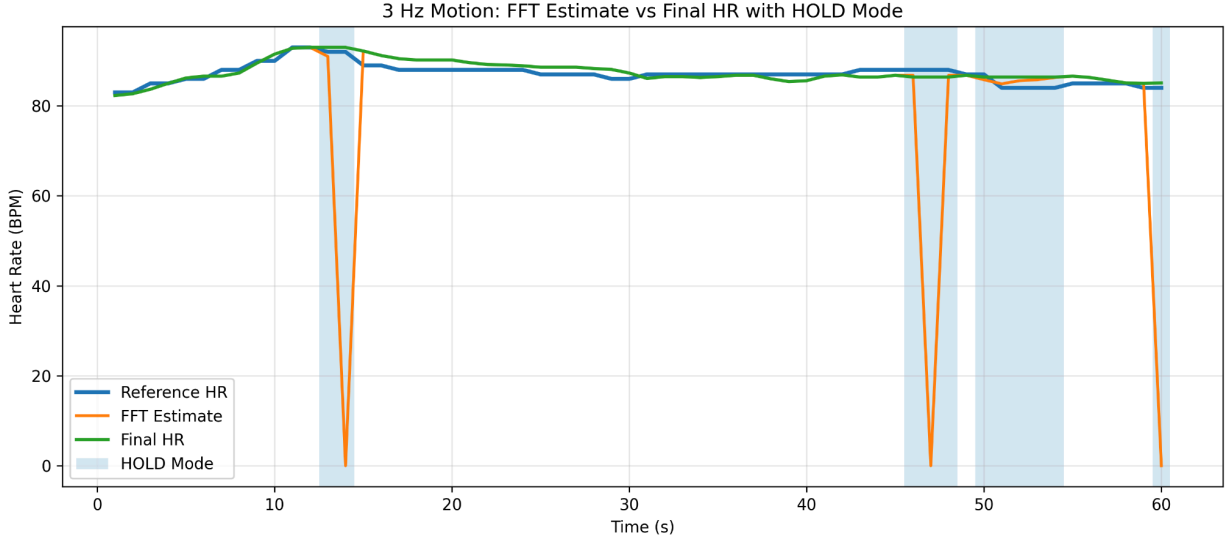


Figure 16: Time-domain comparison of the raw FFT estimate, final heart-rate output, and reference heart rate with hold-mode indication.

The previous section evaluated the overall performance of the proposed framework using the final heart-rate output. To better understand the contribution of the processing stages beyond heart-rate estimation, the raw FFT-based heart-rate estimate was compared with the final heart-rate output under the 3 Hz motion condition, where motion artifacts were most severe.

It is important to distinguish between these two outputs. The FFT estimate represents the direct raw output of Block F, where the dominant frequency component is converted into a heart-rate estimate without any further validation. As a result, this estimate is highly susceptible to motion-induced spectral distortions and may contain large errors when motion artifacts dominate the signal. In contrast, the final heart-rate output is generated after the estimate passes through the complete processing pipeline, including the signal quality assessment and decision layers. The signal quality assessment stage evaluates the reliability of the estimated heart rate, while the decision layer determines whether the estimate should be accepted or rejected. When the signal quality is deemed insufficient, the system enters hold mode and retains the most recently validated heart-rate value instead of updating the output with a potentially erroneous estimate.

Figure 15 compares the error metrics of the raw FFT estimate and the final heart-rate output. The raw FFT estimator produced substantially larger errors, achieving an MAE of 5.48 BPM, an RMSE of

19.73 BPM, and a maximum error of 92 BPM. In comparison, the final heart-rate output achieved an MAE of 1.19 BPM, an RMSE of 1.41 BPM, and a maximum error of only 3.2 BPM. A similar improvement can be observed in the heart-rate standard deviation, which decreased from 19.34 BPM for the FFT estimate to 2.44 BPM for the final output.

The difference between the two outputs is particularly evident in the RMSE and maximum error metrics. As shown in *Figure 16*, the raw FFT estimate occasionally produced extreme outliers due to severe motion corruption. Because the FFT estimator operates solely on the spectral peak detected in Block F, it has no mechanism to determine whether the detected peak corresponds to a true cardiac component or a motion artifact. Consequently, large estimation errors can occur when motion-related frequencies dominate the spectrum. In contrast, these outliers were largely eliminated in the final heart-rate output because unreliable estimates were identified by the signal quality assessment stage and subsequently rejected by the decision layer. During these periods, hold mode prevented erroneous updates by maintaining the last validated heart-rate value until signal quality recovered.

These results highlight the importance of the stages that follow the FFT estimator. While Block F provides an initial heart-rate estimate, the final performance of the system is achieved through the combined action of signal quality assessment, decision making, and hold-mode operation. By preventing unreliable FFT estimates from propagating to the output and maintaining continuity during periods of poor signal quality, the complete pipeline significantly improves both estimation accuracy and output stability under severe motion conditions.

7. Limitations & Future Work

Although the proposed framework demonstrated accurate heart-rate estimation under all tested motion conditions, several limitations remain.

First, the experimental evaluation was conducted using a single participant under controlled hand-swinging motions at frequencies of 1 Hz, 2 Hz, and 3 Hz. While these experiments provided a repeatable method for introducing motion artifacts, they do not fully represent the variety of motion patterns encountered during real-world activities such as walking, running, or daily use. Future work should include testing across multiple participants and a wider range of motion scenarios to further evaluate the generalizability of the proposed framework.

Second, the external reference heart-rate measurements were obtained using a commercial fingertip pulse oximeter and manually recorded at one-second intervals. Although this provided a

practical reference for performance evaluation, a medical-grade electrocardiogram (ECG) system would provide a more accurate ground-truth measurement for future validation.

Third, the adaptive noise cancellation stage utilized only the X-axis acceleration signal as the reference input. Experimental results showed that the X-axis provided the best overall performance among the tested configurations; however, the relationship between motion artifacts and acceleration signals can vary depending on sensor placement, motion direction, and user behavior. Future work may investigate more advanced reference-selection methods or adaptive multi-sensor fusion approaches.

Finally, the results of this study indicate that adaptive noise cancellation alone is insufficient to guarantee robust heart-rate estimation under motion conditions. The largest improvements in performance were achieved through the combined effects of signal preprocessing, frequency-domain heart-rate estimation, signal quality assessment, and hold-mode decision logic. Future work may therefore focus on improving signal quality evaluation, confidence estimation, and motion-aware decision-making strategies rather than relying solely on increasingly complex adaptive filtering techniques.

8. Conclusion

This project presented a motion-aware heart-rate monitoring system that combines PPG and IMU sensing to improve heart-rate estimation under motion conditions. A complete real-time processing pipeline was developed and implemented on an STM32F407 microcontroller, incorporating signal preprocessing, motion metric computation, motion-aware adaptive noise cancellation, FFT-based heart-rate estimation, signal quality assessment, and decision logic.

A key contribution of this work was the introduction of a three-mode NLMS adaptive filtering framework consisting of BYPASS, ADAPT, and FREEZE modes. Experimental results showed that the effectiveness of adaptive filtering depended on motion intensity, with the X-axis acceleration signal providing the most effective reference input under severe motion conditions. However, the results also demonstrated that adaptive filtering alone was insufficient to guarantee robust heart-rate estimation. Under low and moderate motion conditions, the preprocessing stages and FFT-based heart-rate estimation algorithm were already capable of maintaining accurate performance, while adaptive filtering provided the greatest benefit only under higher motion intensities where motion artifacts became more severe.

The results further highlighted the importance of the signal quality assessment and hold-mode framework. By identifying unreliable FFT estimates and preventing them from propagating to the final output, the system significantly reduced large motion-induced estimation errors while maintaining output stability. Under the 3 Hz motion condition, the final heart-rate output reduced the MAE from 5.48 BPM to 1.19 BPM and the RMSE from 19.73 BPM to 1.41 BPM compared with the raw FFT estimate. Overall,

the findings demonstrate that accurate motion-artifact-resistant heart-rate monitoring is achieved not through adaptive filtering alone, but through the combined effects of robust preprocessing, reliable frequency-domain heart-rate estimation, signal quality assessment, and intelligent decision logic operating as an integrated framework.